

Foundations of Research

90 minutes · 5 self-check questions

What separates research from other ways of knowing, the main types of research, and the philosophical positions that quietly shape every methodological choice you will make later.

Learning objectives

- Define research and distinguish it from opinion, audit and routine reporting
- Classify research by purpose, approach and time dimension
- Explain the difference between positivist, interpretivist and pragmatic paradigms
- Trace how a paradigm constrains the designs available to you
- Describe the stages of the research process end to end

What research is, and is not

- Research is systematic, planned enquiry that produces evidence capable of being checked by someone else
- Three features distinguish it: a clear question, a documented method, and conclusions that follow from the data
- It is not: summarising what others said, confirming an opinion you already hold, or collecting data because it is available
- Audit asks "are we meeting an agreed standard?"; research asks "what is the case, and why?"
- The test of research is reproducibility — could a competent stranger follow your method and reach comparable conclusions?

IN PRACTICE

A great deal of published work fails this test. Reviewers increasingly ask for the protocol, the raw data and the analysis code precisely because the written method alone often is not enough to reproduce.

Classifying research

- By purpose: exploratory (what is going on here?), descriptive (what is the pattern?), explanatory (why does it happen?), evaluative (did it work?)
- By approach: quantitative (numbers, measurement, statistical inference), qualitative (meaning, context, interpretation), mixed methods (both, deliberately integrated)

- By time: cross-sectional (a snapshot), longitudinal (repeated over time), retrospective (looking back), prospective (following forward)
- By setting: laboratory, field, clinical, desk-based or secondary analysis
- By application: basic (extends knowledge) versus applied (solves a defined problem)

Paradigms: the assumptions underneath the method

- Ontology asks what reality is: a single objective reality, or multiple realities constructed by participants?
- Epistemology asks how we can know it: by measuring from the outside, or by interpreting from within?
- Positivism assumes an objective reality that can be measured — it leads naturally to experiments, surveys and statistical inference
- Interpretivism assumes reality is socially constructed — it leads to interviews, observation and thematic interpretation
- Pragmatism asks what works for the question at hand, and is the usual justification for mixed methods
- Critical and transformative paradigms foreground power and aim at change, common in participatory and action research

IN PRACTICE

You do not need to win the philosophical argument. You do need your paradigm, question, design and analysis to be consistent with one another — examiners notice when they are not.

The research process

- Identify a problem worth solving and confirm it is not already solved
- Review the literature and locate the gap precisely
- State the question, aims, objectives and (if quantitative) hypotheses
- Choose a design, population, sample and instruments
- Obtain ethical approval before collecting anything
- Collect, manage and analyse data
- Interpret against the literature, then report and disseminate
- The process is iterative in practice — the literature review continues until you submit

Key terms

Paradigm

A set of shared assumptions about the nature of reality and how it can be known, which shapes what counts as legitimate evidence.

Ontology

Assumptions about the nature of reality — whether it is single and objective, or multiple and constructed.

Epistemology

Assumptions about what knowledge is and how it can legitimately be acquired.

Basic research

Research undertaken to extend understanding, without a specific application in view.

Applied research

Research directed at solving an identified practical problem.

Reproducibility

The property that another researcher, following the reported method with the same data, obtains the same result.

Activity

In pairs, take a recent paper from your own field. Identify its purpose, approach, time dimension and implied paradigm. Then argue whether the design actually matches the paradigm the authors claim.

Further reading

- **Creswell & Creswell — Research Design (Ch. 1: The Selection of a Research Approach)** — The clearest short treatment of paradigm-to-design fit.
- **Bryman — Social Research Methods (Ch. 1-2)** — Strong on ontology and epistemology without becoming abstract.
- **Kothari — Research Methodology: Methods and Techniques** — Widely used across African and South Asian universities; good on classification.

Self-check questions

1. A hospital collects readmission rates each quarter and compares them against a national standard to check compliance. This is best described as:
 - a. Explanatory research
 - b. Clinical audit
 - c. Exploratory research

- d. A cohort study
2. A researcher believes patient experience of stigma cannot be captured by a scale because it differs between individuals and settings. Which paradigm is closest to this position?
- a. Positivism
 - b. Interpretivism
 - c. Post-positivism
 - d. Determinism
3. Which sequence correctly orders the research process?
- a. Collect data → review literature → state question → obtain ethical approval
 - b. State question → collect data → obtain ethical approval → analyse
 - c. Review literature → state question → obtain ethical approval → collect data
 - d. Obtain ethical approval → review literature → collect data → state question
4. A study follows a group of 400 newly diagnosed patients forward for three years, measuring outcomes annually. On the time dimension this is:
- a. Cross-sectional
 - b. Retrospective
 - c. Prospective longitudinal
 - d. Case-control
5. The main reason paradigm matters practically is that it:
- a. Determines which journals will accept the paper
 - b. Constrains which designs and forms of evidence are coherent for your question
 - c. Decides whether ethical approval is required
 - d. Sets the required sample size
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Answers

1. **b.** Measuring performance against an agreed external standard is audit. It becomes research only when it asks a question whose answer is not already defined by the standard — for example, why readmission varies between wards.
2. **b.** Treating reality as constructed and specific to person and context is the interpretivist position, which leads towards interviews and observation rather than measurement instruments.
3. **c.** Ethical approval must precede data collection, and the question must be settled before approval can be sought. Collecting first and seeking approval afterwards is a common and serious error that can render data unusable.
4. **c.** Participants are enrolled and then followed forward with repeated measurement, which is prospective and longitudinal. A cross-sectional study would measure once.

5. **b.** A paradigm is not decoration. It governs what counts as evidence, and therefore which designs, instruments and analyses hang together. Mismatches between stated paradigm and actual method are a frequent examiner criticism.